Interstate Migration as a Chord Diagram

Everybody moved, almost nobody left: a year of state-to-state flows in one circle, and the grouping the circle demands

Democracy's Data

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Americans tell the story of their own migration as an exodus. The Northeast empties into the South, California empties into Texas, and whole regions are winners or losers. The story matters to democracy because House seats and electoral votes really do follow the movers. But is the movement actually one-way? Or do the flows mostly run in both directions at once and cancel?

The record that can answer this is a survey. In the year the American Community Survey last measured it, 7,157,607 people in the United States lived in a different state than they had lived in twelve months earlier. Another 29,888,154 moved within their own state, and 2,552,193 arrived from abroad. The between-state moves are published as a matrix: 52 origins by 52 destinations, 2,652 ordered pairs. Each pair carries an estimate and a margin of error, which says how far off that estimate could be.

That matrix is the cleanest flow data in American demography, and it is almost impossible to look at. A **chord diagram** is the form built for exactly this shape of data. It is a circle whose rim is the set of places, with a ribbon joining each pair, wide where the flow is heavy. It will not take 2,652 ribbons. Getting the matrix small enough to draw is most of the work in this brief, and it is where the arguing should happen.

01 The test

The flows are the Census Bureau's *State-to-State Migration Flows* table, from the 2024 American Community Survey one-year estimates. The ACS reaches roughly 3.5 million addresses a year and asks each household where it lived one year ago; that one question is the country's entire measurement of moving. The American Community Survey chapter is the survey's biography: who is in it, how it is weighted up to the country, and why a margin of error stands beside every estimate it publishes.

The numbers here are the same tables the migration brief hands over, suppressed cells and all, so the two documents cannot silently disagree about a number. The test itself is the one the exodus story begs for. Group the states, add up the flow in each direction between every pair of groups, and measure how much is net and how much cancels.

02 The data

One row of the flow table is an ordered pair of states. Direction is a column, not a sign: Middle Atlantic to South Atlantic and South Atlantic to Middle Atlantic are two different rows holding two different numbers, and this brief turns on the fact that both are large.

From	To	People	Note
Middle Atlantic	South Atlantic	307,170	the larger direction
South Atlantic	Middle Atlantic	186,857	the smaller direction

After grouping, the working table keeps six columns:

Column	What it holds	Measurement
from	the group people left	categorical
to	the group they arrived in; when it equals `from`, they changed state without leaving the group	categorical
est	how many people the ACS estimates made that move in the year	count
moe	the margin of error on that estimate, at 90% confidence	continuous
cells	how many state-to-state pairs were added together to make this cell	count
suppressed	how many of those pairs the Census would not release	count

The states are grouped into the Census Bureau's own **nine divisions**, the clusters of neighbouring states — New England, the Middle Atlantic, the Pacific and so on — that the Bureau uses to summarise the country, with Puerto Rico carried as a tenth group.[1] The Bureau's divisions cover only the fifty states and the District of Columbia, and 36,401 people left Puerto Rico for a state that year. The list of which state belongs to which division is the one this book's regional-shift brief uses, checked there against the Bureau's published region totals for every decade since 1910.

[1] The regions and divisions are defined in chapter 6 of the Bureau's *Geographic Areas Reference Manual*, https://www2.census.gov/geo/pdfs/reference/GARM/Ch6GARM.pdf.

Region	Division	States	Population	House seats
Northeast	Middle Atlantic	3	42,020,276	55
Northeast	New England	6	15,250,253	21
Midwest	East North Central	5	47,144,543	62
Midwest	West North Central	7	21,736,206	29
South	East South Central	4	19,706,473	26
South	South Atlantic	9	68,975,950	85
South	West South Central	4	42,594,264	53
West	Mountain	8	25,891,835	33
West	Pacific	5	53,316,668	71
Puerto Rico	Puerto Rico	1	3,188,457	0

The grouping is a decision, not a fact about the country. **The Bureau fixed these divisions in 1910.** They put Delaware and Florida in one group, Maryland in the South, Missouri in the Midwest, and Texas with Arkansas and Oklahoma. Every sentence below about "the South" is a sentence about that list of states, and a different list would draw a different circle from identical data.

Two cleaning decisions travel with the grouping, and both push in a known direction. First, **279 of the 2,652 state pairs are not published at all**: the Bureau releases no estimate when too few sampled households made a move. Aggregation adds them as zeros, because there is nothing else to add, so the group totals run slightly low. `matrix.csv` therefore carries a count of suppressed pairs in every cell, and 78 of the 99 division-to-division cells contain at least one. A cell assembled mostly out of suppressed pairs cannot pass itself off as a measurement.

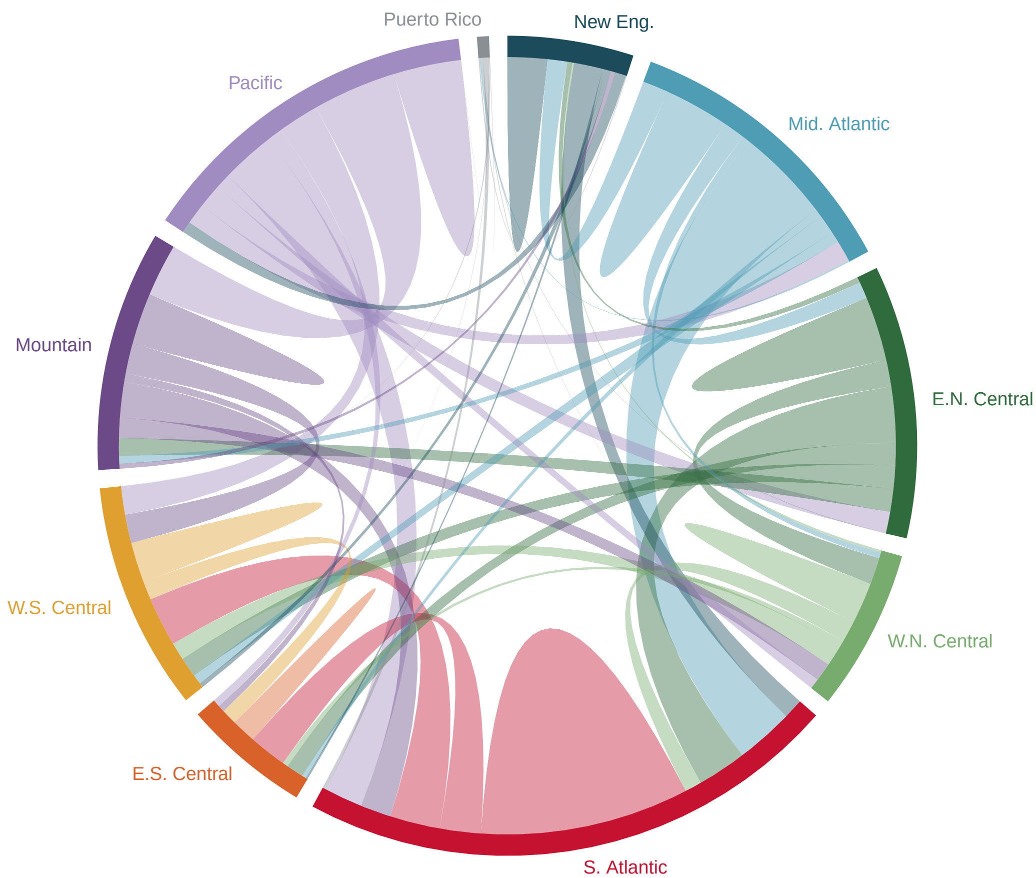
Second, **margins of error on the grouped cells are added in quadrature**, which is the Bureau's own instruction for a sum of published estimates:[2] square each margin, add the squares, and take the square root of the total. The largest exchange in the country shows the arithmetic. Middle Atlantic to South Atlantic carries a margin of ±16,182, and the return flow ±13,094. The net between them is the difference of the two estimates, and its margin is the square root of $16{,}182^2$ plus $13{,}094^2$, which comes to ±20,816 — less than the two added outright, because two errors are unlikely to both fall the same way. That rule assumes the errors are independent, and cells from a single survey are not quite independent. So the margins below are approximate, and if they are wrong they are too small.

[2] U.S. Census Bureau, *Understanding and Using American Community Survey Data: What All Data Users Need to Know* (2020), appendix 3, https://www.census.gov/programs-surveys/acs/library/handbooks/general.html.

> **Before you read on, commit to a number.** Roughly 7.2 million Americans changed state in a year. **What share of them do you think crossed a Census division boundary** — left the Northeast for the South, say — rather than moving to a neighbouring state inside the same division? Write down a percentage.

03 What it says about democracy

Interstate moves, 2024, between and within the nine Census divisions



Ribbon width at each rim is how many people that group SENT; the difference between a ribbon's two ends is the net.

Figure 1. A year of interstate migration, 2024. A chord diagram fits because the data is a square grid of flows among the same ten groups, and the circle is the one form that shows every pair at once. Each rim segment is one division, sized by how many people left it. A ribbon's width where it meets the rim is how many people that division sent, so **the taper of a ribbon is the net flow**; ribbons take the colour of whichever division sent more. Click a rim segment to isolate it; the buttons switch to the four-region grouping and drop the within-division ribbons.

Now the prediction. **26.9% of interstate moves do not cross a division boundary.** Of 7,188,344 moves in the grid, 1,935,522 were to another state in the same division: New York to New Jersey, Virginia to Maryland, Ohio to Michigan. They are the fat lens-shaped ribbons sitting on the rim in Figure 1. The button that hides them is worth pressing twice, because with them gone the circle looks busier and the country looks more mobile than it is.

That is the first thing the grouping does. Every one of those moves was a real change of state, in a matrix built to record changes of state, and grouping turns them into a shape that reads as *staying put*.

The second finding is what the taper of the ribbons is showing, and it is the answer to the exodus story.

Exchange	A to B	B to A	Gross	Net	Net margin	Net share	Clears margin
Middle Atlantic – South Atlantic	307,170	186,857	494,027	120,313	20,816	24.4%	yes
Mountain – Pacific	169,545	270,712	440,257	101,167	18,258	23.0%	yes
East North Central – South Atlantic	179,787	158,068	337,855	21,719	15,812	6.4%	yes
South Atlantic – West South Central	161,793	158,255	320,048	3,538	20,543	1.1%	no
South Atlantic – Pacific	123,341	173,769	297,110	50,428	15,856	17.0%	yes
South Atlantic – East South Central	132,019	121,171	253,190	10,848	15,231	4.3%	no
West South Central – Pacific	91,299	131,305	222,604	40,006	14,087	18.0%	yes
South Atlantic – Mountain	98,753	113,760	212,513	15,007	14,798	7.1%	yes

The largest exchange in the country is Middle Atlantic with South Atlantic: 307,170 people one way, 186,857 back, 494,027 in total and a net of 120,313. That net is the exception that stands. It is 24.4% of the exchange, comfortably clear of its margin, and it is the *most* directional big exchange in the country. Even here, 75.6% of the movement between the two divisions cancelled out.

The typical case cancels harder. **Across all 45 pairs of divisions the median net is 12.1% of the gross flow**, and adding every pair together gives the same answer: 12.6% of all movement between divisions is net. The other 87.4% is two people passing each other in opposite directions. Middle Atlantic and Pacific exchanged 133,689 people for a net of 179, against a margin of 9,506, which is statistically nothing. South Atlantic and West South Central exchanged 320,048 people, a third of a million, for a net of 3,538 with a margin of 20,543.

Only 21 of the 45 division pairs have a net larger than its own margin of error. Group up to the four regions and it is 6 of 10. A net is a difference between two survey estimates and inherits the uncertainty of both, which is why a net of twenty thousand people can be indistinguishable from zero.

Thousands of people, from the row to the column, shaded on a log scale. Grey 0: an estimate of zero, not a missing cell.

	New Eng.	Mid. Atlantic	E.N. Central	W.N. Central	S. Atlantic	E.S. Central	W.S. Central	Mountain	Pacific	Puerto Rico
New Eng.	127	64	17	4	93	9	18	13	37	3
Mid. Atlantic	84	210	68	21	307	26	63	35	67	5
E.N. Central	19	54	205	86	180	72	75	74	67	2
W.N. Central	7	19	85	132	69	19	69	59	33	0
S. Atlantic	64	187	158	54	671	132	162	99	123	11
E.S. Central	6	13	56	18	121	79	51	21	24	0
W.S. Central	14	31	64	57	158	59	129	91	91	0
Mountain	16	25	57	65	114	27	98	164	170	0
Pacific	36	67	77	48	174	49	131	271	217	1
Puerto Rico	3	5	3	3	14	2	4	0	2	

Figure 2. The same matrix, as a matrix. Thousands of people moving from the row's division to the column's, shaded on a log scale, so each step of shading is a tenfold increase rather than a fixed number of people; the diagonal is moves that changed state without leaving the division. Hover any cell for the estimate, its margin of error, and how many suppressed state pairs are inside it. The button switches to net flow, and blanks every pair whose net is smaller than its own margin of error.

Press it. **Most of the matrix goes blank**, and what is left is a small number of genuinely directional exchanges sitting in a field of churn. That is the honest picture of American migration, and no chord diagram can draw it, because a ribbon has no way to say *this taper is not distinguishable from parallel*. The trade runs the other way too: the matrix cannot show at a glance that the Pacific division sends people almost everywhere while East South Central mostly trades with its neighbours. Figure 1 hands you that in a second and will not surrender a single value.

The regions hold one more exception worth telling. The Northeast-to-South net, 200,547 people out of 833,537 exchanged, runs the way the Sun Belt story says. But the South and the West exchanged 1,042,385 people, and their net of 141,743 ran *toward the West*, which is not the direction the story predicts. The button on Figure 1 will show you both.

Why does any of this matter to democracy? Because House seats are apportioned by population, and population moves.

Division	Arrived	Left	Net	Net per 1000	House seats 2020
East South Central	473,482	389,310	+84,172	4.27	26
Mountain	827,891	735,394	+92,497	3.57	33
South Atlantic	1,901,317	1,661,721	+239,596	3.47	85
West South Central	801,009	696,509	+104,500	2.45	53
West North Central	488,769	493,085	-4,316	-0.20	29
New England	377,932	385,241	-7,309	-0.48	21
East North Central	790,360	833,263	-42,903	-0.91	62
Puerto Rico	22,456	36,401	-13,945	-4.37	0
Pacific	830,562	1,070,577	-240,015	-4.50	71
Middle Atlantic	674,566	886,843	-212,277	-5.05	55

The 5 divisions losing people on net hold 238 House seats between them; the 4 gaining hold 197. Middle Atlantic loses 5.0 people per thousand residents a year and East South Central gains 4.3, and a decade of that is what the apportionment brief is recomputing when the seat totals change. The rates are small, and that is the point worth carrying away. Nothing in Figure 1 looks like an exodus, yet a net flow of a few people per thousand per year, sustained for a decade, reorganises the House of Representatives. That is why this table is published annually and read cumulatively.

04 What this brief cannot tell you

Anything the grouping did not decide first. Nine divisions is 99 ribbons; fifty-two units would be 2,652 and unreadable. Every finding above would change under a different list: how much cancels, which pairs stand out, what counts as "within". The Census divisions are a 1910 committee decision, not a natural kind, and a grouping by metropolitan area or by cost of living would draw a different circle from identical numbers.

A trend. This is a single ACS release. The 26.9% of moves that stay inside a division and the 12.1% median net are one year's estimates. The migration brief carries the long series, in which American mobility has been falling for decades.

The suppressed cells. The 279 missing state pairs were added as zeros, and they are the smallest flows between the smallest states, so the rim segments for the least populated divisions are the least trustworthy. They cannot be recovered from the published state totals either, because the ACS estimates its cells and its totals separately and they do not add up.

Why anybody moved, or whether they moved back. Somebody who left Pennsylvania for Florida and returned the next year appears twice, in opposite directions, and contributes nothing to any net. That is not an error in the data; it is what the data is. The word for it is churn, and it is 87.9% of what the median pair of divisions did to each other.

05 What you should have learned

- **26.9% of interstate moves never cross a division boundary**, so a circle drawn from nine groups reads as staying put even though every one of its 7,188,344 moves changed state. Aggregation is a decision with consequences, made before the first pixel is drawn.
- Migration is mostly churn: across the 45 division pairs the median net is 12.1% of the gross flow, and only 12.6% of all between-division movement is net.
- Only 21 of the 45 division nets, and 6 of the 10 region nets, are larger than their own margins of error, because a net inherits the uncertainty of both estimates it is built from.
- The one-way exodus is the exception, not the rule: Middle Atlantic–South Atlantic is the most directional large exchange in the country, and even there 75.6% of the movement cancelled.
- A chord diagram shows the shape of a whole flow grid at a glance and will not surrender a number; the matrix does the opposite. Neither replaces the other.

06 Extensions

- `pairs.csv` has `a_to_b`, `b_to_a`, `gross`, `net` and `net_pct_of_gross`. Sort by the last one. Which pairs are a genuine one-way move, and which are two large flows that almost cancel?
- Count the TRUE rows in the `net_sig` column of `pairs.csv`. What is left of the story about Americans moving between regions once the nets that cannot be told apart from zero are set aside?
- `matrix.csv` records how many state pairs were `suppressed` in each cell. Does suppression fall evenly across the grid, or does it land on particular kinds of place?
- `states.csv` maps each state to both a `division` and a `region`. Regroup the flows by region and compare: does the picture change, and would you have known if you had only ever seen one grouping?
- **Stretch.** Download the previous year's release of the same Bureau table, from the address in the sources below with the year changed, and rebuild the division pairs. Do the same nets clear their margins, or does the set churn?
- **Stretch.** Invent a grouping of your own, by which party carried the state or by cost of living, and reaggregate the state flows in a spreadsheet using your list. Does the exodus story look stronger or weaker than under the Bureau's 1910 map?

07 Sources

Data

- U.S. Census Bureau. *State-to-State Migration Flows*, American Community Survey 2024 one-year estimates, table T13. The figures here read the tables this book's migration brief hands over, which describes the 279 suppressed pairs. https://www2.census.gov/programs-surveys/demo/tables/geographic-mobility/2024/state-to-state-migration/State_to_State_Migration_Table_2024_T13.xlsx
- U.S. Census Bureau. *Geographic Areas Reference Manual*, chapter 6, on the four regions and nine divisions. The state-to-division assignment is read from this book's

regional-shift brief; `states.csv` writes the assignment out again so it can be checked state by state. https://www2.census.gov/geo/pdfs/reference/GARM/Ch6GARM.pdf

- U.S. Census Bureau. *2020 Census Apportionment Results*, as committed by this book's apportionment brief. https://www.census.gov/data/tables/2020/dec/2020-apportionment-data.html
- U.S. Census Bureau (2020). *Understanding and Using American Community Survey Data: What All Data Users Need to Know*, Appendix 3, on aggregating estimates and their margins of error — the source of the quadrature rule used here. https://www.census.gov/programs-surveys/acs/library/handbooks/general.html

Academic literature

- Krzywinski, Martin et al. (2009). "Circos: an information aesthetic for comparative genomics." *Genome Research* 19(9): 1639–1645. The paper that made the chord diagram a standard form, and the source of the layout in Figure 1. https://pmc.ncbi.nlm.nih.gov/articles/PMC2752132/

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`arcs.csv`, `facts.csv`, `groups.csv`, `matrix.csv`, `pairs.csv`, `states.csv`